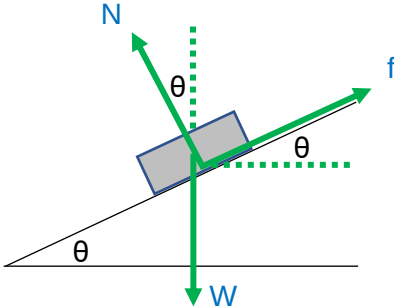


5	(a)	Since the <u>mass is not moving in a straight line</u>, there must be a <u>net force acting</u> on the mass <u>according to Newton's first law</u>. Although the speed remains the same, its direction changes. <u>Since, the change in velocity acts towards the centre of circular motion</u>, the acceleration as well as the net force also acts towards the centre of circular motion <u>according to Newton's second law</u>.	B1 B1
	(b)	 $N \cos \theta + f \sin \theta - W = 0$ $N \cos 40 + 61 \sin 40 - 8.0(9.81) = 0$ $N = 51.26 \text{ N}$ $N = 51 \text{ N}$	M1 A1
	(c)	$f \cos \theta - N \sin \theta = mr\omega^2$ $61 \cos 40 - 51 \sin 40 = (8.0)(12) \left(\frac{2\pi}{T} \right)^2$ $13.9465 = (8.0)(12) \left(\frac{2\pi}{T} \right)^2$ $T = 16.5 \text{ s}$	C1 A1
6	(a)	A field of force (around a mass) is a region of space that can be	B1